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EDUCATION

Muffakham Jah College of Engineering & Technology (Osmania University)	Hyderabad, TG
- Candidate for Bachelor of Engineering in Mechanical Engineering - Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling & Simulation.	May 2028

EXPERIENCE

Department of Mechanical Engineering (MJCET)	
<i>Undergraduate Research Assistant (Edge AI Systems)</i>	December 2025 –Present
- Developed an embedded edge AI platform using an NVIDIA Jetson Nano, IMX719 camera module, and telemetry sensors. - Optimized computer vision inference pipelines using TensorRT FP16 acceleration, achieving 40–70 ms latency (14–20 FPS) under constrained compute environments. - Implemented deterministic execution and monitoring systems for stable long-duration deployment.	
<i>Undergraduate Researcher (Interdepartmental R&D Program)</i>	March 2026 –Present
- Researching adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems. - Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.	
Verity (Decentralized Verification Platform)	Hyderabad, TG
<i>Builder & Systems Architect</i>	July 2025 – Present
- Designing and implementing system to verify digital media provenance using blockchain and decentralized storage - Integrating AI modules for manipulation detection and automated claim extraction	
Verity	Hyderabad, TG
<i>Distributed Systems Developer</i>	July 2025 – Present
- Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave. - Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.	
Freelance Consulting	Hyderabad, TG
<i>Backend Infrastructure Engineer</i>	Sept 2024 – Present
- Migrated client applications to containerized Linux deployment environments with automated CI/CD workflows. - Designed backend APIs and optimized PostgreSQL schemas for improved concurrency and deployment reliability.	

PROJECTS

Smart Agri Four-Legged Bot (SAFL-B)	
<i>Lead Systems Engineer</i>	
- Awarded 1st Place at the MAKEFORHYDERABAD 2-Day Makeathon for autonomous robotics systems integration. - Implemented deterministic UART/Serial communication protocols for multi-joint motor control. - Developed telemetry stabilization routines to reduce terrain-induced sensor and image noise.	
Deterministic Workspace Provisioning	
Built automated provisioning pipelines for lightweight Linux workstation deployment using Arch Linux and shell automation.	

TOOLS & METHODS

Languages: C, Python, Bash, PostgreSQL, Git
Systems: Linux, Arch Linux, Docker, kubernetes, Podman

ML/Edge Computing: TensorRT, NVIDIA Jetson Nano

Hardware & Interfaces: UART/Serial, I2C, GPIO